

- b) The following table gives the data for the duration and costs of each activity of a project network. The indirect cost of project is 3000/week. Determine the optimum duration of the project and the corresponding minimum cost. Draw the time scaled version of the network.

Activity	Normal duration (weeks)	Normal cost	Crash duration (weeks)	Crash cost
1-2	6	7000	3	14500
1-3	8	4000	5	8500
2-3	4	6000	1	9000
2-4	5	8000	3	15000
3-4	5	5000	3	11000

12 L4 4 3
4 L2 4 1

- c) What is the relation between project duration and project cost explain with the help of representation.

8. a) A company manufactures two types of products, P1 and P2. Each product uses a lathe and milling machine. The processing time for P1 on the lathe is five hours and on the milling machine it is four hours. The processing time for P2 on the lathe is 10 hours and on the milling machine it is four hours. The maximum number of hours per week available on the lathe and the milling machine are 60 hours and 40 hours, respectively. The profit per unit of selling P1 and P2 are Rs. 6.00 and Rs. 8.00, respectively. Formulate a linear programming model to determine the production volume of each product such that the total profit is maximized. A company manufactures two types of products, P1 and P2. Each product uses a lathe and milling machine. The processing time for P1 on the lathe is five hours and on the milling machine it is four hours. The processing time for P2 on the lathe is 10 hours and on the milling machine it is four hours. The maximum number of hours per week available on the lathe and the milling machine are 60 hours and 40 hours, respectively. The profit per unit of selling P1 and P2 are Rs. 6.00 and Rs. 8.00, respectively. Formulate a linear programming model to determine the production volume of each product such that the total profit is maximized.

4 L4 4 3

- b) Solve the LPP by simplex method for optimum solution

Minimize $Z = 7x_1 + 5x_2$ subjected to constraints

$$x_1 + 2x_2 \leq 6$$

$$4x_1 + 3x_2 \leq 12$$

$$x_1, x_2 \geq 0$$

8 L5 4 5

- c) Solve the LPP by graphical method

Maximize $Z = 12x_1 + 16x_2$ subjected to constraints

$$10x_1 + 20x_2 \leq 120$$

$$8x_1 + 8x_2 \leq 80$$

$$x_1, x_2 \geq 0$$

8 L4 4 3

Unit – V

9. a) Explain the objectives and functions of material management.
b) What are the basic assumptions of EOQ model?
c) Bob equipment company is in necessity of 48000 units of an equipment parts. Its holding cost is 15% with unit price of Rs.4 and the ordering cost is Rs. 9. Based on the given details suggest how many orders should be placed in a year and how often. Also mention the most economical number of units to order.

8 L2 5 1
4 L1 5 1

8 L3 5 3

22CCT101

SEE-DEC 2024-JAN 2025

10. a) A construction site has limited storage space, what inventory strategies can be implemented to ensure smooth running of project without material shortages or excess?
- b) Explain different Inventory control techniques.
- c) Sarah Brown works for a manufacturer that makes parts for marine engines. The parts are made in batches and every time a new batch is started it costs Rs.1,640 for disruption and lost production and Rs.280 in wages for the fitters. One item has an annual demand of 1,250 units with a selling price of Rs.300, 60% of which is direct material and production costs. If the company looks for a return of 20% a year on capital, what is the optimal batch size for the item and the associated costs?

4	L2	5	2
8	L2	5	1
8	L3	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
